



National
Qualifications
2024

X857/76/22

Physics
Paper 1 — Relationships sheet

THURSDAY, 25 APRIL

9:00 AM – 9:45 AM



* X 8 5 7 7 6 2 2 *

Relationships required for Physics Higher

$$d = \bar{v}t$$

$$s = \bar{v}t$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \frac{1}{2}(u + v)t$$

$$F = ma$$

$$W = mg$$

$$E_w = Fd, \text{ or } W = Fd$$

$$E_p = mgh$$

$$E_k = \frac{1}{2}mv^2$$

$$P = \frac{E}{t}$$

$$p = mv$$

$$Ft = mv - mu$$

$$F = G \frac{m_1 m_2}{r^2}$$

$$t' = \frac{t}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

$$l' = l \sqrt{1 - \left(\frac{v}{c}\right)^2}$$

$$f_o = f_s \left(\frac{v}{v \pm v_s} \right)$$

$$z = \frac{\lambda_{\text{observed}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}}$$

$$z = \frac{v}{c}$$

$$v = H_0 d$$

$$W = QV$$

$$E = mc^2$$

$$I = \frac{P}{A}$$

$$I = \frac{k}{d^2}$$

$$I_1 d_1^2 = I_2 d_2^2$$

$$E = hf$$

$$E_k = hf - hf_0$$

$$v = f\lambda$$

$$E_2 - E_1 = hf$$

$$d \sin \theta = m\lambda$$

$$n = \frac{\sin \theta_1}{\sin \theta_2}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2}$$

$$\sin \theta_c = \frac{1}{n}$$

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}}$$

$$I_{rms} = \frac{I_{peak}}{\sqrt{2}}$$

$$T = \frac{1}{f}$$

$$V = IR$$

$$P = IV = I^2 R = \frac{V^2}{R}$$

$$R_T = R_1 + R_2 + \dots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$V_1 = \left(\frac{R_1}{R_1 + R_2} \right) V_S$$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$E = V + Ir$$

$$C = \frac{Q}{V}$$

$$Q = It$$

$$E = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2} \frac{Q^2}{C}$$

$$\text{path difference} = m\lambda \quad \text{or} \quad \left(m + \frac{1}{2}\right)\lambda \quad \text{where } m = 0, 1, 2, \dots$$

$$\text{random uncertainty} = \frac{\text{max. value} - \text{min. value}}{\text{number of values}}$$

or

$$\Delta R = \frac{R_{\max} - R_{\min}}{n}$$

Additional relationships

Circle

$$\text{circumference} = 2\pi r$$

$$\text{area} = \pi r^2$$

Sphere

$$\text{area} = 4\pi r^2$$

$$\text{volume} = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Electron arrangements of elements

Group 1 Group 2

(1)

Atomic number
Symbol
Electron arrangement
Name

Key

1	H	
1		(2)
Hydrogen		

3	Li	4	Be
2,1		2,2	
Lithium		Beryllium	

11	Na	12	Mg
2,8,1		2,8,2	
Sodium		Magnesium	

19	K	20	Ca
2,8,8,1		2,8,8,2	
Potassium		Calcium	

37	Rb	38	Sr
2,8,18,8,1		2,8,18,8,2	
Rubidium		Strontium	

55	Cs	56	Ba
2,8,18,18,8,1		2,8,18,18,8,2	
Cesium		Barium	

87	Fr	88	Ra
2,8,18,32,18,8,1		2,8,18,32,18,8,2	
Francium		Radium	

Transition elements

(3) (4) (5) (6) (7) (8) (9) (10) (11) (12)

21	Sc	22	Ti	23	V	24	Cr	25	Mn	26	Fe	27	Co	28	Ni	29	Cu	30	Zn
2,8,9,2		2,8,10,2		2,8,11,2		2,8,13,1		2,8,13,2		2,8,14,2		2,8,15,2		2,8,16,2		2,8,18,1		2,8,18,2	
Scandium		Titanium		Vanadium		Chromium		Manganese		Iron		Cobalt		Nickel		Copper		Zinc	

39	Y	40	Zr	41	Nb	42	Mo	43	Tc	44	Ru	45	Rh	46	Pd	47	Ag	48	Cd
2,8,18,9,2		2,8,18,10,2		2,8,18,12,1		2,8,18,13,1		2,8,18,13,2		2,8,18,15,1		2,8,18,16,1		2,8,18,18,0		2,8,18,18,1		2,8,18,18,2	
Yttrium		Zirconium		Niobium		Molybdenum		Technetium		Ruthenium		Rhodium		Palladium		Silver		Cadmium	

57	La	72	Hf	73	Ta	74	W	75	Re	76	Os	77	Ir	78	Pt	79	Au	80	Hg
2,8,18,18,9,2		2,8,18,32,10,2		2,8,18,32,11,2		2,8,18,32,12,2		2,8,18,32,13,2		2,8,18,32,14,2		2,8,18,32,15,2		2,8,18,32,17,1		2,8,18,32,18,1		2,8,18,32,18,2	
Lanthanum		Hafnium		Tantalum		Tungsten		Rhenium		Osmium		Iridium		Platinum		Gold		Mercury	

89	Ac	104	Rf	105	Db	106	Sg	107	Bh	108	Hs	109	Mt	110	Ds	111	Rg	112	Cn
2,8,18,32,18,9,2		2,8,18,32,32,10,2		2,8,18,32,32,11,2		2,8,18,32,32,12,2		2,8,18,32,32,13,2		2,8,18,32,32,14,2		2,8,18,32,32,15,2		2,8,18,32,32,17,1		2,8,18,32,32,18,1		2,8,18,32,32,18,2	
Actinium		Rutherfordium		Dubnium		Seaborgium		Bohrium		Hassium		Meitnerium		Darmstadtium		Koenigsteinium		Copernicium	

2	He	
2		
Helium		

Group 3 Group 4 Group 5 Group 6 Group 7 Group 0

(18)

5	B	6	C	7	N	8	O	9	F	10	Ne
2,3		2,4		2,5		2,6		2,7		2,8	
Boron		Carbon		Nitrogen		Oxygen		Fluorine		Neon	

13	Al	14	Si	15	P	16	S	17	Cl	18	Ar
2,8,3		2,8,4		2,8,5		2,8,6		2,8,7		2,8,8	
Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon	

31	Ga	32	Ge	33	As	34	Se	35	Br	36	Kr
2,8,18,3		2,8,18,4		2,8,18,5		2,8,18,6		2,8,18,7		2,8,18,8	
Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton	

49	In	50	Sn	51	Sb	52	Te	53	I	54	Xe
2,8,18,18,3		2,8,18,18,4		2,8,18,18,5		2,8,18,18,6		2,8,18,18,7		2,8,18,18,8	
Indium		Tin		Antimony		Tellurium		Iodine		Xenon	

81	Tl	82	Pb	83	Bi	84	Po	85	At	86	Rn
2,8,18,32,18,3		2,8,18,32,18,4		2,8,18,32,18,5		2,8,18,32,18,6		2,8,18,32,18,7		2,8,18,32,18,8	
Thallium		Lead		Bismuth		Polonium		Astatine		Radon	

Lanthanides

57	La	58	Ce	59	Pr	60	Nd	61	Pm	62	Sm	63	Eu	64	Gd	65	Tb	66	Dy	67	Ho	68	Er	69	Tm	70	Yb	71	Lu
2,8,18,18,9,2		2,8,18,20,8,2		2,8,18,21,8,2		2,8,18,22,8,2		2,8,18,23,8,2		2,8,18,24,8,2		2,8,18,25,8,2		2,8,18,25,9,2		2,8,18,27,8,2		2,8,18,28,8,2		2,8,18,29,8,2		2,8,18,30,8,2		2,8,18,31,8,2		2,8,18,32,8,2		2,8,18,32,9,2	
Lanthanum		Cerium		Praseodymium		Neodymium		Promethium		Samarium		Europium		Gadolinium		Terbium		Dysprosium		Holmium		Erbium		Thulium		Ytterbium		Lutetium	

Actinides

89	Ac	90	Th	91	Pa	92	U	93	Np	94	Pu	95	Am	96	Cm	97	Bk	98	Cf	99	Es	100	Fm	101	Md	102	No	103	Lr
2,8,18,32,18,9,2		2,8,18,32,18,10,2		2,8,18,32,20,9,2		2,8,18,32,21,9,2		2,8,18,32,22,9,2		2,8,18,32,24,8,2		2,8,18,32,25,8,2		2,8,18,32,25,9,2		2,8,18,32,27,8,2		2,8,18,32,28,8,2		2,8,18,32,29,8,2		2,8,18,32,30,8,2		2,8,18,32,31,8,2		2,8,18,32,32,8,2		2,8,18,32,32,9,2	
Actinium		Thorium		Protactinium		Uranium		Neptunium		Plutonium		Americium		Curium		Berkelium		Californium		Einsteinium		Fermium		Mendelevium		Nobelium		Lawrencium	